WORKSHEET

Geometric probability distributions

1 Two dice are rolled until a pair of sixes is observed. What is the probability that exactly two rolls are required? Circle the correct answer.

A $\frac{35}{36}$

B $\frac{1225}{46\,656}$

C $\frac{35}{1296}$

D $\frac{1}{1296}$

E $\frac{1}{36}$

2 A random variable, Z, represents the number of failures before the first success in a series of independent Bernoulli trials. If the probability of success in each trial is 68%, what is $P(Z = 3)$? Circle the correct answer.

A 0.0328

B 0.0223

C 0.6528

D 0.0504

E 0.2138

3 Determine if the following are examples of geometric distributions. Circle the correct answer.

a A coin is flipped and the number of tails is counted until the first head appears. Yes/No

b The time it takes to complete an ultra-marathon. Yes/No

c The number of heads recorded after flipping a coin three times. Yes/No

d The number of times a die is rolled before a score of six appears. Yes/No

e The number of components inspected before a faulty one is found. Yes/No

f The number of times a score of one is recorded after rolling a die 5 times. Yes/No

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4 An unbiased coin is flipped and the number of heads is counted until a tail is observed. Find the probability that the number of heads is:

a 0

b 1

c 2

d 5

e 20

f 100

5 Random variable, X, has a geometric distribution where X is the number of failures before the first success. If the probability of failure is 0.1, find:

a $P(X = 0)$

b $P(X = 2)$

c $P(X = 5)$

d $P(X = 8)$

e $P(X \leq 2)$

f $P(X > 4)$

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6 A card is randomly drawn from a standard 52 card deck. It is noted whether the card is a spade before it is returned to the deck and the process repeated. If Y is the number of cards drawn before the first spade appears, find:

a $P(Y = 1)$

b $P(Y = 3)$

c $P(Y < 2)$

d $P(Y \geq 5)$

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Answers

1 C

2 B

3 **a** Yes

b No

c No

d Yes

e Yes

f No

4 **a** 0.5

b 0.25

c 0.125

d ≈ 0.0156

e $\approx 4.768 \times 10^{-7}$

f $\approx 3.944 \times 10^{-31}$

5 **a** 0.9

b 0.009

c 9×10^{-6}

d 9×10^{-9}

e 0.999

f 1×10^{-5}

6 **a** $\frac{3}{16}$

b $\frac{27}{256}$

c $\frac{7}{16}$

d $\frac{243}{1024}$

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